

3 The protein content of food can be estimated using a semi-quantitative test.

(a) (i) Name a reagent that can produce a semi-quantitative test for protein.

(1)

(ii) State the colour change that occurs when this reagent is added to a protein solution.

(1)

(iii) A semi-quantitative test was used to estimate the protein concentration in a solution.

A set of standards was produced using dilutions of egg white containing protein of known concentrations.

The test was carried out on each standard and on the sample with an unknown protein concentration (test tube 6).

The photograph shows the results.

						
Test tube	1	2	3	4	5	6
Egg white protein concentration / mg dm^{-3}	0	0.1	0.4	2.0	10.0	unknown

(Source: © Martyn F. Chillmaid / Science Photo Library)

Estimate the protein concentration in the solution in test tube 6.

Express your answer as a range.

(1)



- (iv) Suggest how this investigation could be adapted to increase the accuracy of the estimate.

(2)

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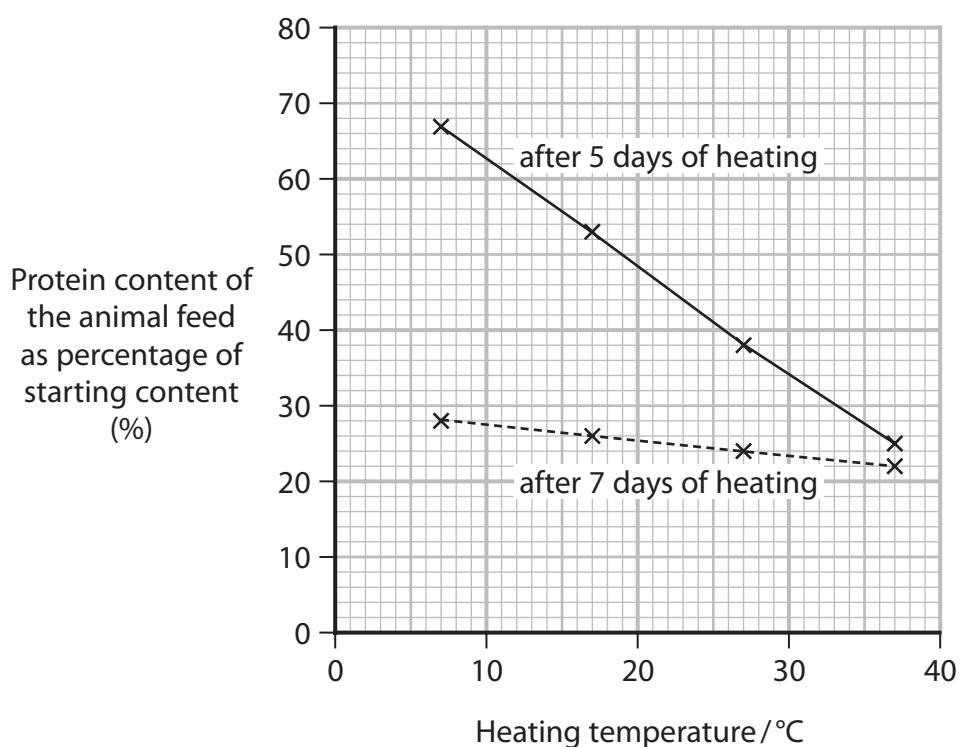
- (b) Some herring are processed to make animal feed.
The photograph shows a catch of herring.



(Source: © John Greim / Science Photo Library)

This process adds acid to the fish followed by heating them for a period of time. The effect of temperature and heating time on the protein content of the animal feed was investigated.

The graph shows the protein content after heating for 5 or 7 days at each temperature.



(i) Draw a table to display all the results shown in this graph.

(3)

(ii) Calculate the gradient of the line at 7 days.

Include units in your answer.

(3)

Answer

(iii) Comment on the effects of both heating temperature and the length of time heated on the protein content of the animal feed.

Use the graph and your calculation in (b)(ii) to support your answer.

(3)

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- (iv) This process preserves the animal feed because it kills bacteria.
The process involves heating for a period of time, the heating temperature and the addition of acid.

Devise a procedure to safely compare the effect of acid on bacteria.

(6)

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(Total for Question 3 = 20 marks)

TOTAL FOR PAPER = 50 MARKS

